Dangerous Viable Space (DVS): A Spatial Control Framework for Quantifying NFL Wide Receiver Route-Running Threat

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Abstract

In the NFL, route-running is one of a wide receiver's (WR) most important skills. While current NFL wide receiver route-running metrics are often rudimentary and lack information about the threat of a given route, our research quantifies route-running ability by combining spatial control and expected threat. By applying Spearman's soccer pitch control methodology (2017) to NFL tracking data, we created a physics-based, machine-learning-less model to identify WR and defensive back (DB) spatial control over the course of a pass play. To better understand the value of the space WRs control relative to DBs, we weight it by an xEPA model that assigns a threat value to each field location. This product yields the WR's Dangerous Viable Space (DVS) at each point in time. By taking the average and maximum of the wide receiver's DVS, we can calculate both the receiver's Average Route Threat Efficiency (aRTE) and Maximum Route Threat Efficiency (mRTE), respectively. By isolating wide receiver route-running ability from target share and throw outcome, field control, aRTE, and mRTE offer clearer metrics for evaluating downfield threat, scheme fit, and receiver archetypes.

1 Introduction & Literature Review

In American football, evaluating a wide receiver's (WR) route-running performance remains heavily polluted by downstream outcomes and lacks comprehensive metrics that evaluate all components of route running. Traditional box-score metrics, such as targets, yards per route run, and completion percentage, are fundamentally conditional on quarterback decision-making, offensive line protection, and play design. Modern Next Gen Stats tracking metrics evaluate receivers only at discrete static moments, such as separation at the time of ball arrival (NFL Next Gen Stats, n.d.). This raw separation misses the complete spatial picture; a receiver might have significant separation at release, only for a defender's closing speed to erase that gap almost immediately.

This creates a critical analytical gap: a receiver who runs a crisp route and establishes leverage in high-value space receives zero or little credit if the quarterback progresses elsewhere. Meanwhile, a receiver who wins late against a blown coverage can accumulate inflated outcome metrics. To truly isolate route-running quality from team-level noise, an evaluation framework must continuously track spatial control, weight that space by its contextual scoring value, and bound the evaluation to physically realistic pass trajectories.

To capture the contextual value of a route independently of whether it results in a catch, we build upon the Expected Points (EP) framework. Originally developed to estimate the "next score" from a given down, distance, and yard line (Carter and Machol 1971), EP evolved into Expected Points Added (EPA) to quantify the value of individual plays (Burke 2010) and isolate individual player contributions to team success (Yurko, Ventura, and Horowitz 2019). Our research extends this idea spatially, assigning a localized expected drive-value (xEPA) to every point on the grid based on historical passing data.

To measure who actually controls those valuable locations, we adapt physics-based pitch control, originally developed to estimate soccer pass completion probabilities using player time-to-intercept (Spearman 2017). In soccer analytics, researchers have successfully combined this field control with expected goals (Spearman 2018; Fernandez and Bornn 2018) and developed metrics like Dangerous Accessible Space (Bischofberger and Baca 2025) to identify players generating valuable spatial opportunities.

American football research has recently adopted pitch control to evaluate quarterback pressure (Inayatali, White, and Hocevar 2023) and quantify defensive pass coverage (Subbaiah, Chu, Reyers, and Wu 2021). While prior football research like Illuminating the Defense (Subbaiah et al., 2021) successfully weighted field control by applying a direct linear multiplication of spatial xEPA and pitch control, these applications have historically focused on defensive evaluation. No existing NFL framework combines spatial control with a threat-weighted field value specifically to isolate a wide receiver's downfield route-running threat.

To fill this gap, we introduce Dangerous Viable Space (DVS). Instead of direct linear multiplication, DVS uses a non-linear transformation of xEPA and pitch control to better capture the risk-reward dynamics of NFL passing windows. Crucially,

we restrict this evaluation exclusively to Reasonable Pass Bounds: a dynamically calculated Pareto frontier that isolates physically viable throw envelopes given player kinematics and throw geometry. By accumulating DVS across each frame of a route, we propose three novel, target-independent evaluation metrics:

- **Average Route Threat Efficiency (aRTE)**: Quantifies a receiver's ability to sustain control over high-value space throughout the duration of a route.
- **Maximum Route Threat Efficiency (mRTE)**: Captures peak window-creation ability, measuring how effectively a receiver creates maximum threat during key timing windows.
- **Final Route Threat Efficiency (fRTE)**: Captures a receiver's ability to sustain a downfield threat at the exact moment of the throw.

By stripping out target bias, play outcome variance, and screen-play execution noise, aRTE, mRTE, and fRTE provide coaches and front-office analysts with objective metrics for evaluating isolated WR skill, scheme fit, and receiver archetypes.

The remainder of this paper is structured as follows: Section 2 outlines the NFL Big Data Bowl dataset and our data pipeline; Section 3 details the spatial xEPA field valuation models, the adapted physics-based pitch control model, and the Pareto-frontier pass bounds; Section 4 presents empirical results and validates the RTE metrics across player archetypes; and Section 5 discusses practical applications for NFL front offices and coaching staff.

2 Data

Data for this study are sourced from the open-access 2026 NFL Big Data Bowl dataset (NFL, 2025), comprising over 14,000 passing attempts from the 2023 NFL regular season. The dataset combines high-frequency spatial-temporal tracking captured at 10 Hz (10 observations per second) with rich play-level metadata. The tracking data records continuous positional coordinates (x, y), speed, acceleration, and orientation for all eligible receivers, defensive backs, and quarterbacks from the moment the ball is snapped until the pass is rendered complete or incomplete.

To ensure physical and spatial consistency across all plays, raw tracking coordinates were transformed into a standardized left-to-right offensive coordinate system ($x \in [0, 120]$, $y \in [0, 53.33]$ yards). This spatial normalization enforces directional invariance across kinematic vectors—such as velocity ($\vec{v}$), acceleration ($\vec{a}$), and orientation (θ)—mapping every route onto a uniform 120×54 one-yard cellular grid regardless of drive direction or end zone orientation.

Play-level metadata (down, distance, yardline, and baseline expected points) was joined with frame-level tracking via unique relational identifiers (gameId, playId, nflId). To systematically evaluate route execution, the temporal window for each play is partitioned into two distinct phases:

- **Pre-pass phase**: From ball snap to pass release, representing the core analytical window for evaluating continuous route-running threat generation.
- **Post-pass phase**: From pass release to ball arrival or outcome, utilized primarily to calibrate pass trajectory bounds and completion probabilities.

Plays designated as wide receiver screens, spikes, sacks, or pre-pass penalties were filtered out of the primary sample to isolate true downfield route execution from deliberate backfield design and play-breakdown noise.

3 Methodology & Modeling Framework

3.1 Field Control Model

Our pitch control framework adapts William Spearman's (2017) physics-based spatial ownership model to the specific kinematics of American football. Using purely player positioning and velocity, we are able to capture pass completion probabilities at each location on the field. Because the flight profile and arrival mechanics of an aerial football pass differ fundamentally from a soccer pass, we re-engineered the ball kinematics to map expected completion percentages continuously across the field.

To isolate individual wide receiver route-running ability, the model evaluates a single targeted receiver against all secondary defenders on a uniform 120×54 grid of 1-yard by 1-yard cells. This approach strips away the noise of offensive scheme and supporting cast positioning to isolate a given WR's value creation.

Kinematics and Time-to-Intercept (TTI)

Player movement toward any field coordinate is partitioned into an acceleration phase and a cruising phase. Current speed (v_0) is extracted from tracking data, while maximum velocity (v_{max}) and maximum acceleration (a_{max}) are dynamically bounded

using each player's historical 99th-percentile tracking metrics. The distance a player can cover during their acceleration phase (t_{acc}) is calculated as $d_{acc} = v_0 t_{acc} + \frac{1}{2} a_{max} t_{acc}^2$.

Each player's Time-to-Intercept (TTI) is computed based on whether they will reach maximum velocity before reaching the target point.

If the target distance $d \leq d_{acc}$, the Time-to-Intercept (TTI) is solved via the quadratic equation:

$$TTI = \frac{-v_0 + \sqrt{v_0^2 + 2a_{max} d}}{a_{max}}$$

If the player reaches maximum velocity before arriving at the target, then the remainder of the distance is traversed at constant speed:

$$TTI = t_{acc} + \frac{d - d_{acc}}{v_{max}}$$

Ball flight times are determined by calculating the distance to the target and bounding the 2D velocity of the football (v_{ball}) between 14 yds/sec (lob passes) and 23 yds/sec (bullet passes), reflecting empirical 5th and 95th percentile NFL throw speeds. 2D velocity is captured because the data is limited to x and y coordinates; it is not feasible to measure the arc of the ball without a vertical z coordinate. In multi-receiver computations, the ball arrival time is calibrated to the receiver who arrives first, assuming optimal quarterback execution. By assuming optimal quarterback execution, we strip away the effects of offensive line play-to-play blocking, the quarterback's ability to navigate the pocket and potentially throw on the run depending on the situation. By not considering these effects, we are better able to isolate the receiver's route-running ability and the threat they create in a given play.

Spatial Control and Arrival Probability

The probability of a player arriving at a target cell by time t uses a logistic arrival model based on the difference between the current time and their TTI:

$$P(t) = \frac{1}{1 + e^{-c(t - TTI)}}$$

(where c represents the logistic scaling variance constant). Field control is calculated iteratively across infinitesimally small time intervals (dt). As players approach the target, their exerted control accumulates. To govern this dynamic control surface, we enforce the following football-specific mechanical assumptions:

- **Arrival-Dependent Control & Decay**: Player control over any field location remains strictly zero until the simulated pass reaches that location ($t < T_{\text{ball}}$). Once the ball arrives ($t \geq T_{\text{ball}}$), control accumulates dynamically over time step dt. As total accumulated control ($PPCF_{\text{tot}}$) approaches 100%, the available unallocated space ($(1 - PPCF_{\text{tot}})$) diminishes, making it exponentially harder for opposing players to usurp an already controlled cell.The incremental change in Pitch Control ($dPPCF$) is modeled as:

$$dPPCF = \begin{cases} 0, & t < T_{\text{ball}} \\ (1 - PPCF_{\text{tot}}) \cdot (P(t) \cdot \lambda) \cdot dt, & t \geq T_{\text{ball}} \end{cases}$$

  where T_{ball} is the time required for the football to reach the target cell, $P(t)$ is the player's logistic arrival probability, and λ is their base control rate.

- **Defensive Advantage (λ)**: Following Spearman's (2018) foundational pitch control framework, the defensive control rate (λ_{def}) is scaled by a coefficient of $\kappa = 1.18$. Derived via Maximum Likelihood Estimation, this grants defenders a 17% spatial acquisition advantage, mathematically reflecting the empirical reality that disrupting a pass requires significantly less time and precision than catching and securing one. It is important to note that while Spearman's paper derived pitch control parameters for soccer, this framework uses coefficients optimized for football.

- **Forward Pass Masking**: Expected completion percentage is strictly masked to 0 for all spatial coordinates behind the quarterback ($x \leq x_{qb}$). Backward passes mechanically function as run plays. Allowing receivers to accumulate spatial value behind the line of scrimmage would distort the evaluation of downfield route-running threat.

- **Quarterback Exclusion**: The quarterback's spatial control rate (λ_{qb}) is permanently forced to 0. The quarterback cannot complete a pass to himself, so his tracking coordinates only serve as the geometric origin point for throw distance calculations.

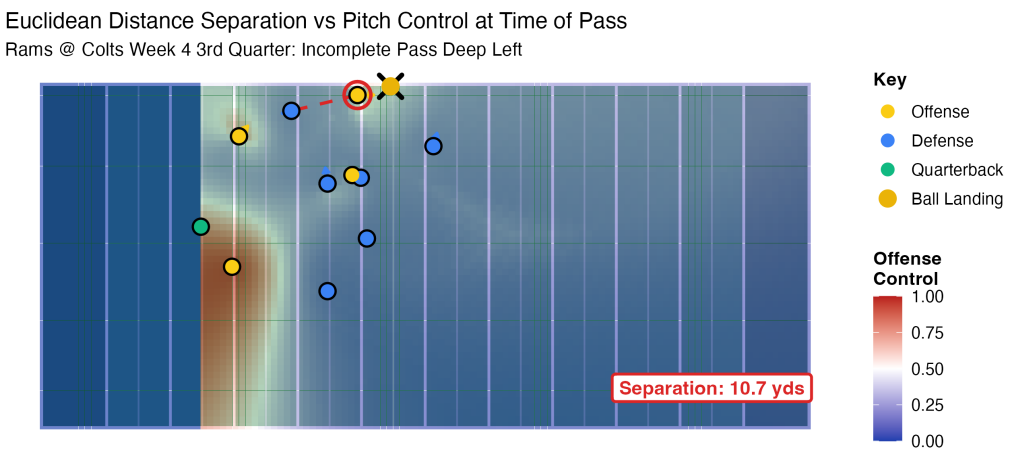


Figure 1: Field control visualization example.

Figure 1 visualizes raw field control during a 2023 Week 4 Rams vs. Colts play. Despite high separation at the time of the throw, the receiver lacks control over the surrounding area, resulting in an incomplete pass. This illustrates how the field control mechanism captures continuous spatial dynamics, unlike traditional separation metrics that are limited to a discrete point in time.

3.2 Field Value Model – Completions ($xEPA_c$)

The field value model quantifies the magnitude of drive-specific Expected Points (EP) generated at each point on the field, conditional on the pass being completed, by isolating the specific context of a play independent of game-state noise like win probability. A 20-yard reception on 3rd & 15 carries identical intrinsic spatial value whether a team is tied or trailing by 30 points. To evaluate wide receivers independent of their supporting cast and overarching game state, Expected Points isolations best answer where and how much field position gains support a team's localized effort to score.

Data Architecture and Spatial Symmetry

The response variable for the model is Post-Play Expected Points given a completion ($PostEP_c$), calculated by adding the realized Expected Points Added given a completion (EPA_c) to the pre-play Expected Points ($PreEP$). To prevent the model from learning artificial directional biases based on empirical dataset tendencies (e.g., right-handed quarterbacks preferring to throw to the right sideline), coordinate geometry is forced into strict symmetry. Prior to fitting, the spatial training data is duplicated and reflected across the field's longitudinal center. Every empirical pass coordinate is appended with a mirrored counterpart by transforming the vertical coordinate as $y_{mirrored} = 53.3 - y_{std}$.

Generalized Additive Model (GAM) Formulation

To map continuous expected value across the field, the framework utilizes a Generalized Additive Model (GAM). The expected post-play value is modeled using a 3-dimensional tensor product smooth for the spatial parameters, while strictly controlling for pre-play expected points via a separate 1D marginal smooth.

The full model takes the form:

$$\mathbb{E}[PostEP_c] = \beta_0 + f_{spatial}(x_{depth}, y_{land}, x_{los}) + f_{context}(PreEP)$$

Where: - $f_{spatial}$ is a tensor product (te) constructed from marginal splines with basis dimension $k = 8$ for pass depth (x_{depth}), transverse field location (y_{land}), and line of scrimmage (x_{los}). Pass depth is dynamically calculated relative to the line of scrimmage: $x_{depth} = x_{land} - x_{los}$.
- $f_{context}$ is a one-dimensional smooth (s) of the pre-play $PreEP$ with basis dimension $k = 15$.

To robustly handle heavy-tailed residuals generated by structural edge cases (specifically low-frequency, explosive touchdown plays where the localized EP strictly jumps to 6) the model is fit using a Scaled t-distribution (scat) with an identity link function. Utilizing a heavy-tailed distribution prevents these extreme scoring outliers from aggressively pulling the spline bases and distorting the general route-value surface.

Yield and Expected Points Added ($xEPA_c$)

The continuous expected value surface is generated by interpolating predictions across a 0.5-yard resolution grid. The model actively bounds the evaluation space by masking out impossible spatial geometries, dropping any coordinates that land beyond the back of the endzone ($x_{los} + x_{depth} \leq 110$).

To isolate the expected spatial Expected Points Added ($xEPA_c$) at any specific grid coordinate, the $PreEP$ of the play context is subtracted from the model's predicted $PostEP$:

$$xEPA_c(x, y) = \mathbb{E}[PostEP_c] - PreEP$$

By combining this continuous $xEPA$ surface with the kinematic pitch control matrices, the expected value of a receiver's spatial control can be evaluated on a frame-by-frame basis throughout the route.

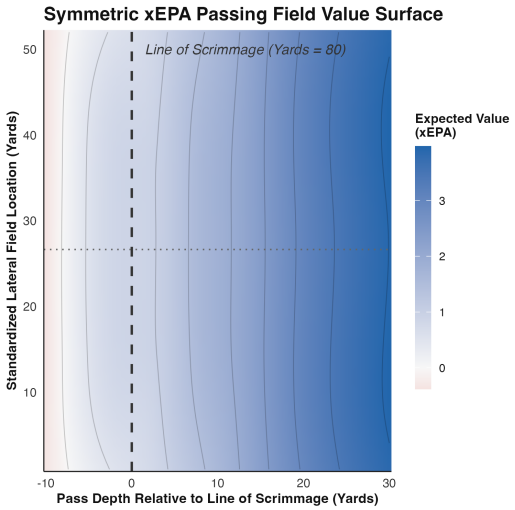


Figure 2: xEPA Field Value Example

Figure 2 visualizes the xEPA for completed pass plays starting from the opponent's 30 yardline. Intuitively, the xEPA increases gradually as the pass length increases, and horizontal change does not materially impact the xEPA of a pass.

3.3 Field Value Model – Incompletions (ΔEPA_{inc})

A completion is only one of the two ways a throw can end, so the value of a receiver's space cannot be judged by the completion outcome alone. If a receiver controls a cell that is worth a lot when caught but is rarely won, the cost of the pass falling incomplete should count against it. The incompletion model quantifies that cost as the change in Expected Points (EP) when the pass is not completed.

Play-Level Value of an Incompletion

Unlike the completion value in Section 3.2, the value of an incompletion does not depend on where the ball lands. An incomplete pass returns the offense to the line of scrimmage with a new down, so the outcome depends only on the game state at the snap: down, distance, and field position. For that reason ΔEPA_{inc} is a single number for each play that is applied identically to every cell of the grid. It is defined as the difference between the EP of the state after the incompletion and the EP of the current state:

$$\Delta EPA_{inc} = EP(\text{state after incompletion}) - EP(\text{current state})$$

The state after an incompletion depends on the down:

- **1st through 3rd down**: The ball stays at the same spot with the same distance to go, and the down advances by one.
- **4th down**: A turnover on downs. The defense takes over at the spot of the snap at 1st down with 10 yards to go (goal-to-go if the spot is inside the 10), and the value to the offense is the negative of that state's EP. With yl as the offense's distance to the goal line, this is $EP_{next} = -EP(1, \min(10, 100 - yl), 100 - yl)$.

Expected Points State Model

Both terms of the difference require an EP estimate for an arbitrary combination of down, distance, and field position, including hypothetical states like 4th & 7 that did not happen in the play at hand. We therefore fit a separate GAM to the

play-level data, with the play's pre-snap expected points as the response. Separate smooths of yardline and yards to go are fit within each down:

$$\mathbb{E}[EP] = \beta_0 + \beta_{down} + f_{down}(yl) + g_{down}(yd)$$

Where yl is the distance to the opponent's goal line (110 minus the line of scrimmage on our 120-yard field, bounded between 1 and 99) and yd is yards to go, which is bounded to be at least 1 and no more than the distance to the goal line or 25 yards. The yardline smooths use basis dimension $k = 8$, the yards-to-go smooths use $k = 5$, and the model is fit with REML.

To check the model, we use 5-fold cross validation with folds grouped by game, so that no play from a held-out game is ever used to train the model that predicts it. Out-of-fold accuracy (RMSE, MAE, and R^2) is compared to a baseline that predicts the mean EP of each down. Because only differences between two EP predictions enter ΔEPA_{inc}, any overall bias in the model cancels out, and what matters is that the model orders states correctly: an incompletion should almost always cost the offense expected points. We check this directly by computing the share of plays on each down for which the implied ΔEPA_{inc} is negative. One limitation is that the 3rd-to-4th down step is partially an extrapolation, since 4th downs in the data are mostly ones where the team chose to go for it and are therefore concentrated at short distances.

3.4 Reasonable Pass Bounds

To prevent evaluating defensive spatial control over physically unviable pass trajectories, the modeling evaluation space is constrained to a dynamically calculated Pareto boundary called the Reasonable Pass Bounds. This spatial envelope establishes the domain of realistic pass landing locations given the target receiver's route trajectory, velocity vector, and lookahead horizon at any given time.

We use a Pareto distribution, rather than a simple Gaussian, to model the heavy right-tailed distribution of pass landing locations. In our residual analysis, we found an extreme right tail that doesn't reflect a normal distribution, highlighting the importance of our design choice to use a Pareto distribution for this model.

Sample Filtering and Dynamic Categorization

The calibration pipeline strictly isolates completed passes exclusively during the pre-pass phase. This filter was chosen to identify pass locations that had a reasonable chance of being completed. For instance, we deliberately remove throwaways, overthrown deep passes, and batted balls. Pass depth is dynamically calculated relative to the line of scrimmage utilizing the receiver's anchor coordinates at the exact throw frame (the maximum available pre-pass frame):

$$x_{\text{depth}} = x_{\text{anchor}} - x_{\text{los}}$$

Rather than utilizing static yardage cutoffs, route depths are partitioned, using a simple classifier model, into three discrete bins based on the sample mean (μ) and standard deviation (σ) of the empirical release depths: Short ($< \mu - \sigma$), Medium ($[\mu - \sigma, \mu + \sigma]$), and Deep ($> \mu + \sigma$).

Route Geometry and Spatial Envelope Mechanics

Our data has pre-pass movement for all receivers, but is limited to just post-pass locations for the targeted receiver. Therefore, one of the parameters we seek to optimize is a lookahead horizon that accounts for the receiver's movement at the time of the pass, allowing us to map the receiver's future, post-pass position whether they were targeted or not. For a given lookahead horizon of N frames post-current frame ($N \in \{5, 10, \ldots, 30\}$), the receiver's step speed (v) and movement unit vector are computed using their current frame diagnostics. A projected forward segment length (L) is extrapolated from the receiver's anchor point $(x_{\text{anchor}}, y_{\text{anchor}})$ by multiplying the step speed by the lookahead frame count:

$$L = v \times N$$

The total spatial bounds around this trajectory are modeled as the union of two geometric regions:

- **Forward Corridor**: A tubular buffer extending along the projected route path with a lateral half-width w. A coordinate falls within the forward tube if its orthogonal distance to the projected segment satisfies $d_{\perp} \leq w$, and its projected arc length along the path satisfies $0 \leq s \leq L$.

- **Backward Cushion Zone**: To account for underthrown adjustments, a rectangular cushion extends behind the throw anchor point along the inverse of the receiver's initial direction vector $\vec{u}$. A point falls in this backward zone if its longitudinal backward displacement satisfies $0 \leq d_{\parallel,\text{back}} \leq b$ and its perpendicular offset satisfies $d_{\perp,\text{back}} \leq w$.

- **Corridor Width (w)**: The lateral half-width of the forward tube and backward cushion zone, which is optimized independently for each depth bin to account for kinematic variations across route depths.

A pass coordinate is enclosed within the total spatial envelope if it falls into either the forward tube or the backward cushion zone. These limits are computed for each frame of the WR's route pre-throw.

Stratified Grid Calibration and Optimization

Parameters are optimized independently for each depth bin k to account for kinematic variations across route depths utilizing a parallelized grid search. The search space evaluates backward cushions $b \in [0, 3]$ yards (step 0.5), corridor widths $w \in [1, 6]$ yards (step 0.5), and lookahead horizons N tailored specifically to each depth bin. For each depth bin, the optimal parameter tuple (b_k^*, w_k^*, N_k^*) minimizes the total envelope surface area ($Area$) while guaranteeing a target empirical pass coverage threshold $\alpha \geq 0.90$:

$$\min_{w,b,N} \quad 2w(L+b) + \pi w^2 \quad \text{subject to} \quad \frac{1}{M_k} \sum_{i=1}^{M_k} \mathbb{I}_{\text{inside},i} \geq 0.90$$

where M_k is the total play count in bin k. The 90% minimum threshold was selected to strike an optimal balance: it captures the vast majority of historically realistic target locations while explicitly filtering out anomalous outliers. These outliers include severely deflected passes, egregious miscommunications, or highly improbable desperation throws that would otherwise artificially inflate the spatial envelope and dilute the accuracy of the defensive evaluation.

To maintain physical consistency, corridor widths are constrained to be strictly monotonic across increasing depth bins utilizing a cumulative maximum function. This monotonic restriction ensures that deeper passes will be treated with more flexible reasonable pass bounds due to increased ball time in air and time for the receiver to reach the ball. Ultimately, our calibrated parameters successfully bound the target zones, yielding final empirical coverage rates of 90.9% for Short routes, 90.2% for Medium routes, and 85% for Deep routes. The results were also tested on held out data that was not used for optimization and our model achieved very similar results on the test set.

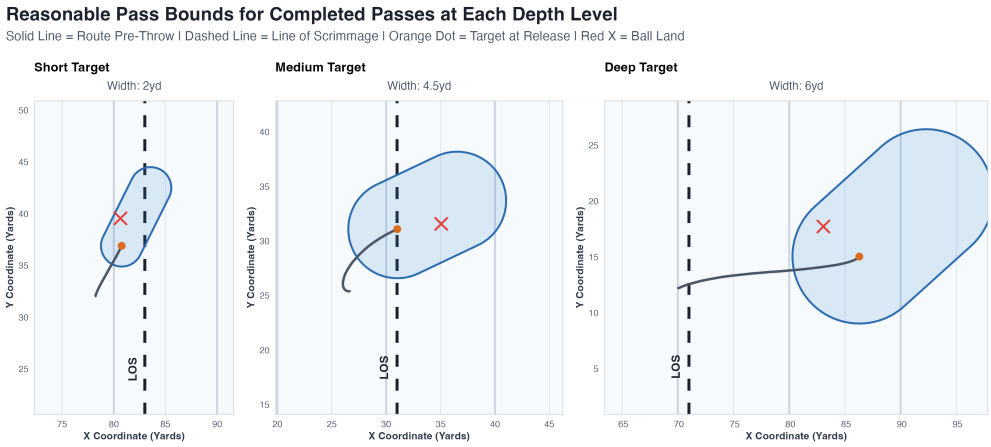


Figure 3: Reasonable Pass Bounds Examples

As illustrated in Figure 3, the resulting spatial envelopes dynamically scale with target depth, expanding from a tight 2-yard width on Short targets to 4.5 yards on Medium targets, and reaching 6 yards on Deep throws. In each representative case, the actual pass landing location (red "X") falls cleanly within the boundary calibrated from the receiver's pre-throw trajectory (solid line) and position at release (orange dot). By appropriately expanding the corridor as route depth and air time increase, this approach successfully isolates physically viable target zones without overestimating the receiver's reachable area.

Residual Diagnostics

Spatial residual errors are decomposed into perpendicular lateral errors ($r_\perp = d_\perp$) and longitudinal depth errors relative to the projected line segment ($r_\parallel = s - L$). Diagnostic evaluation confirms that the vast majority of longitudinal residuals are negative, demonstrating that the projected forward segment acts as an effective upper bound that successfully encloses all catchable targets.

Variance across route depths relative to the line of scrimmage is assessed using non-parametric LOESS smoothing. The smoothing demonstrates general stability across most route depths, although localized variance and structural shifts emerge within the 10-to-20-yard intermediate range. These shifts are due to the use of discrete bucketing, wherein routes that barely cross into the next depth threshold may have larger than realistic bounds since they fall in the shallower end of a deep pass bucket.

The overall distribution of lateral errors is evaluated utilizing density histograms and normal Q-Q plots of model residuals. Rather than confirming normality, these diagnostics reveal a heavily right-skewed distribution; the distribution exhibits a hard lower bound near zero and a significant upward deviation from theoretical normal quantiles, reflecting a long tail of extreme lateral outliers. This heavy tail supports the use of empirical thresholds via grid search rather than linear models to set parameters for reasonable pass bounds.

3.5 Dangerous Viable Space (DVS)

Dangerous Viable Space (DVS) represents the mathematical synthesis of the kinematic field control model and the continuous expected points surface. To calculate the localized threat generated by a receiver at any discrete field coordinate (x, y), the framework computes a combined metric integrating spatial ownership with the intrinsic expected scoring value of that specific location.

This framework forgoes the direct linear multiplication of spatial Expected Points Added ($xEPA$) and offensive Pitch Control Probability ($PPCF_{att}$) utilized in prior models such as Illuminating the Defense (Subbaiah et al., 2021). Instead, the model applies a specific non-linear transformation to better capture the risk-reward dynamics of NFL passing windows. The raw spatial efficiency at each grid cell is modeled as:

$$p(x, y) = PPCF_{att}(x, y)$$
$$DVS(x, y) = p \cdot xEPA_c + (1 - p)^2 \cdot 0.75 \cdot \Delta EPA_{inc}$$

Here p is the weight placed on the completion outcome, and the remaining $(1 - p)$ is placed on the incompletion outcome, whose value ΔEPA_{inc} (Section 3.3) is the same for every cell on a given play. Squaring the defensive control probability $(1 - p)$ and scaling down the incompletion penalty shifts additional weight to the offense, whose control is slightly dampened by the model due to the defensive advantage adjustment mentioned earlier. This transformation strictly rewards receivers who establish dominant leverage in positive-value areas rather than receivers who are in high-value locations with little spatial control. To ensure the metric strictly evaluates downfield route-running threat, DVS is subjected to rigid spatial masking. Any calculated efficiency value located behind the quarterback's dropback position is zeroed out, preventing lateral or backward movement behind the line of scrimmage from artificially deflating a receiver's spatial score.

With the continuous DVS surface established, these cellular values are aggregated to evaluate route execution dynamically across time. For any given frame t during the pre-pass phase, the receiver's aggregate Frame Efficiency (Frame DVS) is calculated by summing the DVS values across all valid spatial cells within the targeted evaluation zone:

$$FrameDVS_t = \sum_{(x,y) \in \text{Target Zone}} DVS(x, y, t)$$

From this continuous array of frame-level threat generation, the model extracts three primary play-level evaluation metrics to quantify route-running proficiency independent of quarterback targets:

- **Average Route Threat Efficiency (aRTE)**: Calculated as the mean Frame DVS across all analyzed pre-pass frames. This metric quantifies a receiver's baseline ability to sustain high-value spatial control throughout the entire duration of a route's development.
- **Maximum Route Threat Efficiency (mRTE)**: Isolated as the peak (maximum) Frame DVS achieved at any point during the pre-pass lifecycle. This captures a receiver's peak window-creation ability, measuring their capacity to generate explosive threat at the optimal timing of a play
- **Final Route Threat Efficiency (fRTE)**: Computed as the Frame DVS at the time of throw. This reflects a receiver's route-finishing capability, evaluating how much spatial control and drive-value potential they maintain at the exact window of opportunity.

Employing these same aggregations over only $PPCF_{att}$ yields the following metrics. They serve the same purpose as their RTE conterparts while only accounting for separation:

- **Average Offensive Field Control (aOFC)**
- **Maximum Offensive Field Control (mOFC)**
- **Final Offensive Field Control (fOFC)**

For these final metrics, it's important to acknowledge that we are not controlling for the temporal aspect of a play. In other words, for longer plays receivers have more chances (frames) to generate a high maximum threat, raise their average, and open themselves up for a higher final threat or field control value. However, by analyzing these over our route depth classes, we are able to control for the fact that longer plays allow receivers to get further down field and have more time to generate threat.

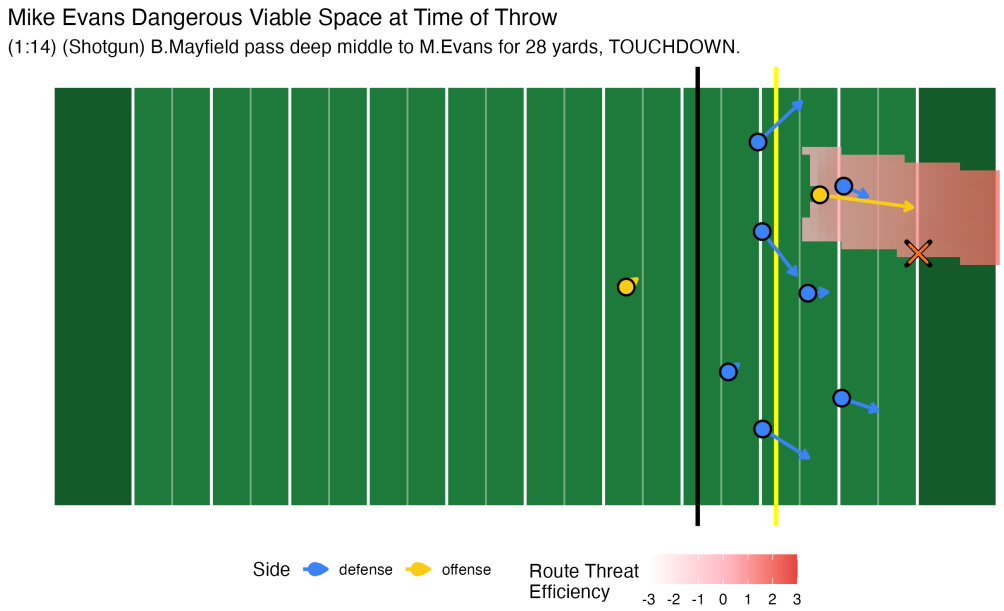


Figure 4: Dangerous Viable Space (DVS) Example

Figure 4 illustrates this framework in action during a 28-yard touchdown pass to Mike Evans. At the exact frame of pass release, corresponding to the Final Route Threat Efficiency (fRTE) metric, Evans' route leverage and speed vector (yellow dot and arrow) establish a dominant, high-efficiency corridor (shaded red) extending well past the line of scrimmage (black line) and the line to gain (yellow line). Notably, the actual ball landing spot (indicated by the "X") falls directly within the highest concentration of threat efficiency, demonstrating how Evans effectively cleared defensive leverage to create valuable targetable space.

4 Analysis & Empirical Results

We ran the full pipeline on every wide receiver in the 2023 season and asked two questions of each metric: is it stable, and does it tell us anything about what happens next? Alongside RTE we test its field-control-only counterparts (aOFC, mOFC, and fOFC, which are the same aggregations from Section 3.5 applied to $PPCF_{att}$ alone), traditional separation (the distance to the nearest defender at the time of the throw), and, where we need an outcome, air yards per route, completion percentage, and EPA per target. Each receiver's value is the median across his routes in a route group. Receivers are grouped into Short, Medium, and Long routes; Ultra-Long routes are left out because too few receivers run enough of them.

4.1 Stability

To test stability, we split the 2023 season at its league-wide midpoint and correlated each receiver's first-half value with his second-half value, requiring at least 20 routes of the route group in each half. Figure 5 shows the results.

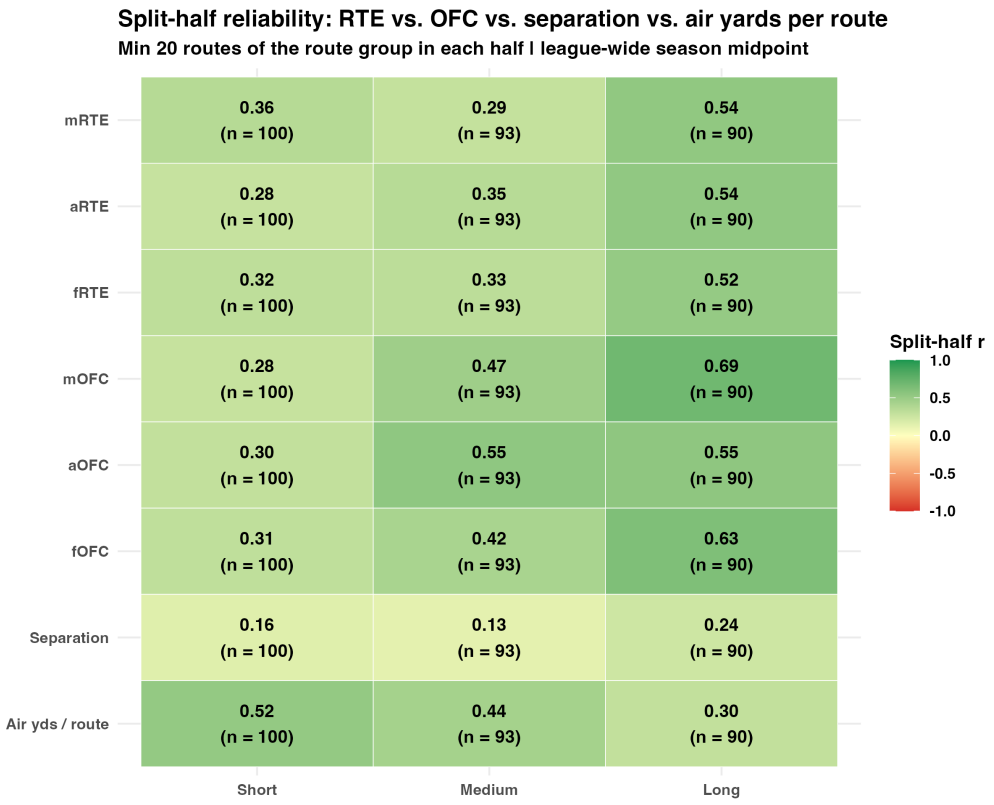


Figure 5: Split-half reliability of RTE, field control, separation, and air yards per route by route group.

Separation is the clear outlier. Its split-half correlation sits between 0.13 and 0.24 in every route group, so how much separation a receiver creates in the first half says very little about the second. Everything we built holds up better, albeit with a small sample size. Field control is the most stable family on medium and long routes: aOFC is 0.55 in both, and on long routes mOFC and fOFC reach 0.69 and 0.63, well above the 0.30 for air yards per route. RTE is a step lower but still reliable, and peak RTE is the standout on short routes at 0.65, ahead of air yards per route (0.52). Short-route field control is the weak spot for the OFC family at around 0.30. This is the case in favor of tracking a whole route instead of a single frame: a snapshot at the throw is noisy, while an entire route's worth of frames is not.

4.2 What They Predict

Stable doesn't mean useful, so we next asked whether first-half values predict second-half outcomes: air yards per route, completion percentage, and EPA per target. We judge each feature by its leave-one-receiver-out skill, which is how much better it predicts a held-out receiver than a baseline with no feature (above 0 is better). We also include a receiver's own first-half result as a benchmark.

The short answer is not much. A receiver's own first-half air yards per route is the best predictor of his second-half air yards per route (skill of about 0.20 on short and medium routes, 0.07 on long), and none of separation, field control, or RTE adds signal beyond it. Completion percentage and EPA per target are harder targets, since they only exist on targeted routes and EPA carries quarterback and yards-after-catch noise, and there too nothing meaningfully beats a baseline with no feature. We found three exceptions:

- **Short-route completion percentage**: Peak field control has a small edge, with a correlation of 0.199 with second-half completion percentage. That is as close to significant as anything gets here ($p = 0.094$), but out-of-sample skill barely moves (0.013 for mOFC).
- **Long-route air yards per route**: aRTE has a relatively strong negative correlation to second-half air yards per route ($r = -0.298$). This does provide some out-of-sample skill (+0.072) with significance ($p = 0.004$). After further analysis, this negative correlation and signal can be strongly attributed to a lower target rate on deeper routes with higher RTE. Corr(aRTE, Target Rate) = -0.60.

- **Long-route EPA per target**: Separation is a positive indicator here (skill of 0.061, $r = 0.286$, $p = 0.017$), the one place it beats the field.

Taken together, these metrics measure something real and repeatable about a receiver's route running, but not something

that shows up as extra production or efficiency.

4.3 Receiver Archetypes

Where the metrics earn their keep is describing how a receiver wins. Figure 6 plots each receiver's peak field control (mOFC) against his average (aOFC). Receivers above the dotted line have a peak that outruns their average: they create their space in sharp, brief windows, and we call them route runners. Receivers below the line sustain their field control rather than spike it, and we call them speed separators.

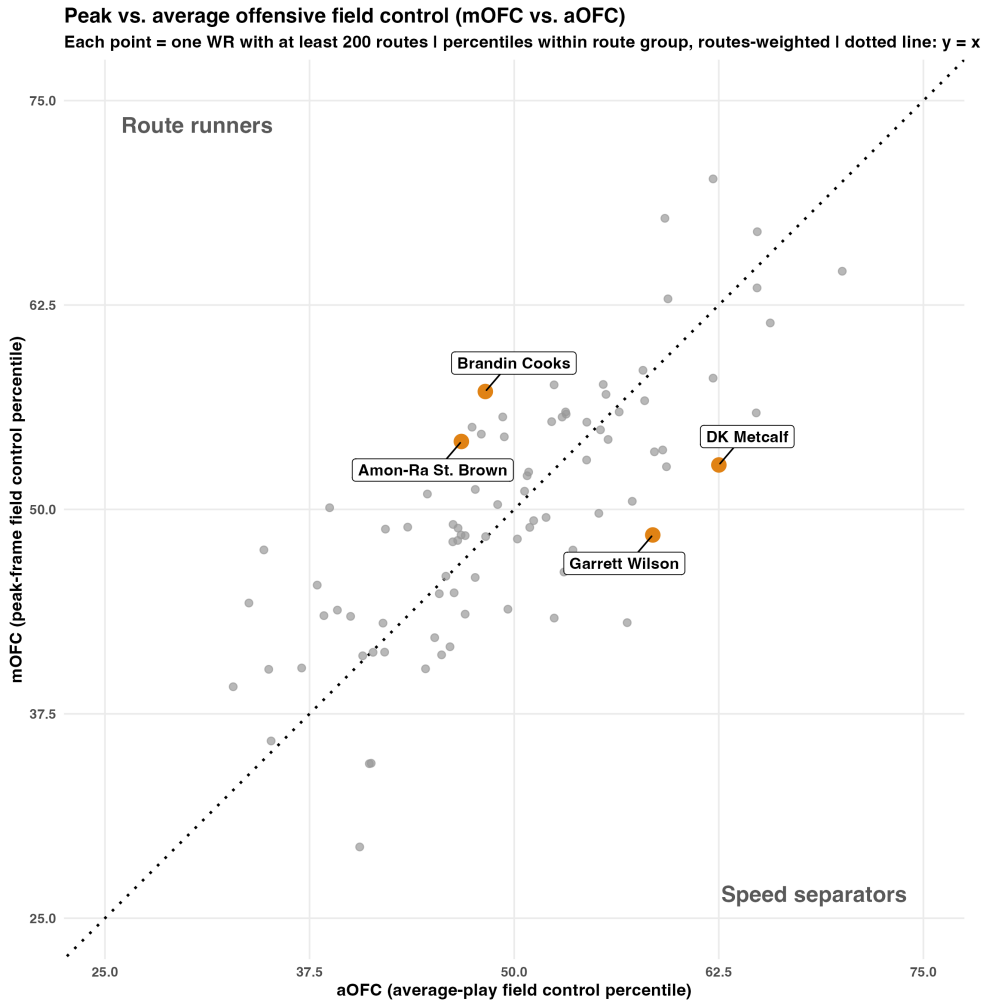


Figure 6: Peak versus average field control (mOFC vs. aOFC) for receivers with at least 200 routes.

The highlighted names land about where you would expect: Brandin Cooks and Amon-Ra St. Brown above the line, DK Metcalf and Garrett Wilson below it. The same split works with mRTE and aRTE. It also looks like how far a receiver sits above or below the line, meaning how much his peak over- or under-performs his average, may be a marker of latent route-running talent. That is a hypothesis, not a result, and we haven't tested it here.

5 NFL Applications

Given what we found, the metrics are best used to describe how a receiver gets open, not to forecast how much he will produce. They should sit beside production metrics, not replace them.

5.1 Front Office & Roster Construction

- **Archetype Profiling**: Teams can pair complementary skill sets by identifying route runners (peak above average) and speed separators (average above peak), and check whether a receiver's profile fits the scheme he would be asked to run.
- **Scouting on Small Samples**: The metrics were stable with only 20 routes in each half of a season, and they don't depend on whether the quarterback looked his way. A receiver's profile can be read without waiting for target volume.

5.2 Coaching Staff & Tactical Game Planning

- **Pass Timing & Concept Design**: Frame-by-frame DVS shows when in a route the window opens, which coaches can use to line up a quarterback's progression with a receiver's peak and to see which concepts create space.
- **Defensive Leverage**: Defensive coordinators can invert the framework to find receivers who create the most space at the moment of the throw ($fRTE$) and bracket them.

6 Discussion & Limitations

6.1 Analytical Contributions

This research advances football analytics by synthesizing kinematic pitch control with localized expected drive value ($xEPA$). The DVS framework introduces a non-linear threat transformation that aggressively penalizes contested space, restricts spatial evaluation to physically viable "Reasonable Pass Bounds," and establishes target-independent metrics to accurately project how and where receivers get open.

6.2 Limitations & Future Research

- **Depth vs. Separation**: Because deeper routes naturally retain a mathematical advantage, future iterations should incorporate more aggressive adaptive depth normalization to better balance short-area separators against vertical threats.
- **Multi-Receiver Dynamics**: The current model evaluates isolated WR-DB interactions. Expanding to a multi-agent model would help quantify a receiver's "distraction value" in clearing out coverage for teammates.
- **3D Kinematics**: Lacking z-axis data limits the evaluation of ball height and player elevation. Incorporating vertical tracking will refine time-to-intercept models for high-point and contested catches.

Ultimately, the Dangerous Viable Space framework redefines how wide receiver performance is measured by decoupling individual route-running execution from team-level outcomes. By continuously tracking spatial threat generation rather than relying on static, target-dependent results, the derived RTE and OFC metrics provide a more objective and robust foundation for both roster construction and tactical game planning in the modern NFL.

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